

ECG Simulator Documentation

Part 7: Technical Reference & Validation

Numerical Methods, CFL Stability, Performance Benchmarks

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1 CFL Stability Analysis

1.1 Finite-Difference Discretization

For the 1-D monodomain cable:

$$\frac{\partial v}{\partial t} = D \frac{\partial^2 v}{\partial x^2} + I_{\text{ion}}(v, w) \quad (1)$$

Explicit Euler in space (central difference) and time:

$$v_i^{n+1} = v_i^n + \Delta t \left[D \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{(\Delta x)^2} + I_{\text{ion}} \right] \quad (2)$$

1.2 CFL (Courant–Friedrichs–Lewy) Condition

For stability of explicit Euler:

$$\text{CFL} = \frac{D\Delta t}{(\Delta x)^2} \leq 0.5 \quad (3)$$

1.3 Validation for Phase 3 Cable

Region	D (cm ² /s)	Δx (cm)	CFL	Status	
Atrium (RA/LA)	0.288	0.2	0.192		
Purkinje (LBB)	7.2	0.2	0.48		
LV Myocardium	0.30	0.2	0.02		
Max CFL			0.48	Safe	

All regions satisfy CFL 0.5. Explicit Euler is stable for chosen parameters.

2 Numerical Integration Parameters

2.1 Phase 3 Cable Integration

Parameter	Value	Purpose	
τ_s (time scale)	0.015 s	Physical time normalization	
Δt_{int} (micro step)	0.0001 s	Integration step	
Δt_{frame} (display)	0.033 s	GUI update (30 FPS)	
N_{micro} (steps/frame)	$\text{ceil}(0.033 / 0.0001) = 330$	Steps per display update	

2.2 Phase 3-D Calibration

Numerical search for τ_s matching clinical ECG timings:

τ_s (ms)	P-wave (ms)	PR (ms)	QRS (ms)	
13	—	—	— (CFL violated)	
15	64	196	111 (SELECTED)	
20	104	266	151	
25	191	435	210	

Selected $\tau_s = 15$ ms matches all clinical targets.

3 Clinical Validation Data

3.1 Nominal ECG Parameters (Normal Sinus Rhythm)

Parameter	Simulated	Clinical Normal	Status	
HR	70 bpm	60–100 bpm		
P-wave duration	64 ms	40–100 ms		
PR interval	196 ms	120–200 ms		
QRS duration	111 ms	80–120 ms		
QT interval	420 ms	360–440 ms (male)		

3.2 Amplitude Validation

Wave	Simulated	Clinical Normal	Status	
QRS amplitude	1.287 mV	0.5–2.5 mV		
T-wave amplitude	0.590 mV	0.1–0.6 mV		
QRS/T ratio	2.18	1.5–3.0		

4 Performance Benchmarks

4.1 Computational Speed

Engine	Speed	Notes	
Phase 1/2 (BFS)	~1000 Hz	GPU-limited by display refresh	
Phase 3 (FHN Cable)	2.2× real-time	Continuous real-time capability	
Display refresh	30 FPS	PyQtGraph update rate	

4.2 Multi-Beat Performance

At 70 bpm (HR) with Phase 3 cable:

- **6 beats in 5 seconds:** Confirmed
- **Emergent refractoriness:** FHN recovery time $\tau_w \approx 234$ ms
- **Real-time display:** Smooth scrolling, no lag

5 Installation & Dependencies

5.1 System Requirements

- **OS:** Linux, macOS, or Windows
- **Python:** 3.10 or newer
- **RAM:** 512 MB minimum, 2 GB recommended
- **Disk:** 100 MB for source code + dependencies

5.2 Python Dependencies

Package	Version	Purpose	
PyQt6	6.0	GUI framework	
NumPy	2.0	Numerical computations	
pyqtgraph	0.12	Real-time ECG plotting	
scipy	1.10	Optional: advanced integration	

5.3 Installation Command

```
pip install PyQt6>=6.0 numpy>=2.0 pyqtgraph>=0.12
```

5.4 Running the Simulator

```
cd /path/to/ECG_simulator
python main.py
```

6 Glossary of Terms

Term	Definition
Action Potential (AP)	Rapid, temporary change in electrical potential of a cell
Automaticity	Ability of cardiac tissue to spontaneously depolarize
AV Node	Tissue between atria and ventricles; delays impulse
Bundle Branch	Part of conduction system in ventricles
CFL Condition	Courant–Friedrichs–Lewy stability criterion for explicit methods
Conduction Velocity	Speed at which electrical impulse travels through tissue
Depolarization	Change from resting potential to positive values
ECG/EKG	Electrocardiogram; electrical recording of heart
Ectopic Focus	Abnormal site of electrical activity
Excitability	Ease with which tissue can be stimulated
Fibrillation	Rapid, disorganized electrical activity
FitzHugh-Nagumo	Simplified 2-variable cell model
His Bundle	Part of conduction system; junction of AV node and bundle branches
Hyperpolarization	Potential more negative than resting (recovery phase)
Inactivation	State where ion channels cannot reopen immediately
Ischemia	Reduced blood flow to tissue
Lead Vector	Direction of electrical observation for 12-lead ECG
Monodomain	Mathematical model of continuous cardiac tissue
Refractoriness	Period when tissue cannot be stimulated
Repolarization	Return to resting potential
SA Node	Sinoatrial node; natural pacemaker of heart

7 References

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